

## BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

### Standard Operating Procedure

SOP No: BSP-CC-003

Title: Breakout Response at Slab Caster

Department: Continuous Casting

Revision: 03

Date: 22-Aug-2024

#### 1. PURPOSE

To define immediate actions in case of a breakout (liquid steel escaping through the solidified shell) at the slab casters.

#### 2. SCOPE

Applies to Slab Caster Nos. 1 and 2, including casting floor, segment area and run-out table personnel.

#### 3. EARLY WARNING SIGNS

- Breakout Prediction System (BOPS) alarm on the HMI
- Sudden rise in mould thermocouple temperature (more than 15°C in 10 seconds)
- Abnormal mould level fluctuation above  $\pm 5$  mm
- Sparks or flames observed below the mould

#### 4. IMMEDIATE ACTIONS ON BOPS ALARM

Step 1 — Reduce casting speed to 0.4 m/min automatically or manually.

Step 2 — Hold reduced speed for at least 30 seconds and watch thermocouple trends.

Step 3 — If the alarm clears, increase speed gradually by 0.1 m/min every minute.

#### 5. ACTIONS ON CONFIRMED BREAKOUT

Step 1 — Press the EMERGENCY STOP on the casting floor; the tundish slide gate closes automatically.

Step 2 — Close the ladle shroud and swing the ladle turret to the safe position.

Step 3 — Evacuate the segment area and run-out table; nobody may stay below the casting floor.

Step 4 — Keep secondary cooling water ON to protect segments unless instructed otherwise.

Step 5 — Inform Shift In-charge (Ext. 3310), Emergency Control Room (Ext. 100) and Maintenance (Ext. 3325).

#### 6. PPE FOR RECOVERY WORK

- Aluminised suit, face shield, heat-resistant gloves and safety boots
- Recovery work only under a valid Hot Work Permit (see BSP-SAF-014)

#### 7. REPORTING

Breakouts must be reported in the Caster Incident Register within 2 hours.